

Supplementary Material: Detailed Benchmark Results

This supplement reports detailed benchmark results for all datasets and both labelling strategies (`mvc`, `classic`). Each cell shows mean $\pm$ standard deviation over 10 seeds. Columns labelled 0.2, 0.5, and 0.8 refer to target labelling rates $c_{\text{calc}} \in \{0.2, 0.5, 0.8\}$.

1 Benchmark Overview

Table S2 summarizes the best observed F1 score for each dataset and labelling strategy, together with the Naive baseline and the absolute gain. These numbers are the basis for the quantitative statements in the abstract and conclusion.

Table S1 clarifies how the current manuscript extends the earlier non-SCAR study.

Table S1: Comparison of the our earlier non-SCAR study and the present manuscript.

Aspect	Earlier study	Present manuscript
Task framing	Narrower non-SCAR benchmark centered on clustering and logistic modeling	Broader PU benchmark with six methods and two non-SCAR strategies
Implementation	Initial pipeline and fewer implementation details	Python reimplementations with reproducible preprocessing and evaluation
Experimental scope	Smaller method comparison	Six SpeakLeash corpora, three c_{calc} values, ten seeds
Interpretation	Focus on model behavior	Adds dataset-level discussion, limitations, and practical selection guidance
Reporting	Summary of benchmark behavior	Expanded runtime analysis, stability analysis, and best-F1 summary

Table S2: Best observed F1 score per dataset and strategy, together with the Naive baseline and absolute gain.

Dataset	Strategy	Best F1	Naive F1	Gain
ISAP_corpus	classic	0.019	0.000	0.019
job_offers_pl_corpus	classic	0.613	0.014	0.599
open_subtitles_corpus	classic	0.906	0.861	0.045
plwiki	classic	0.771	0.013	0.758
ulotki_medyczne	classic	0.833	0.022	0.811
wolne_lektury_corpus	classic	0.692	0.365	0.327
ISAP_corpus	mvc	0.020	0.000	0.020
job_offers_pl_corpus	mvc	0.339	0.004	0.335
open_subtitles_corpus	mvc	0.900	0.827	0.073
plwiki	mvc	0.716	0.045	0.671
ulotki_medyczne	mvc	0.209	0.000	0.209
wolne_lektury_corpus	mvc	0.573	0.074	0.499

Table S3: Average wall-clock runtime by method and labelling strategy, aggregated over datasets and target labelling rates.

Method	classic [min]	mvc [min]
Naive	0.003	0.003
Clust	0.026	0.029
Strict-LassClust	4.347	3.809
Non-Strict-LassClust	4.426	4.220
LassoJoint	2.086	2.011
Spy	0.028	0.027

Table S4: Practical method-selection guide derived from the benchmark results.

Data condition	Preferred method(s)	Reason
Moderately balanced corpus with clear feature signal	LassoJoint	Strongest average retrieval quality and good use of sparse informative features
Highly imbalanced corpus or weak propensity signal	Spy	More stable when positive support is small and class separation is difficult
Need for a lightweight baseline or quick sanity check	Clust / Naive	Low runtime and simple behavior make them useful reference points
Need to stress-test label-generation assumptions	Compare classic vs mvc	Strategy sensitivity is dataset-specific and should be interpreted as part of the benchmark design

2 Labelling Strategy: mvc

Table S5: Detailed results for **ISAP_corpus** with labelling strategy **mvc** (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.089 $\pm$ 0.017	0.127 $\pm$ 0.018	0.249 $\pm$ 0.025	0.005	0.005	0.005	0.000	0.000	0.000
Clust	0.772 $\pm$ 0.043	0.104 $\pm$ 0.025	0.184 $\pm$ 0.030	0.022 $\pm$ 0.003	0.005	0.005	0.017	0.000	0.000
S-LassClust	0.820 $\pm$ 0.051	0.090 $\pm$ 0.021	0.144 $\pm$ 0.023	0.036$\pm$0.008	0.005	0.005	0.017	0.000	0.000
NS-LassClust	0.821$\pm$0.050	0.090 $\pm$ 0.021	0.144 $\pm$ 0.023	0.036$\pm$0.008	0.005	0.005	0.017	0.000	0.000
LassoJoint	0.144 $\pm$ 0.036	0.176 $\pm$ 0.021	0.241 $\pm$ 0.009	0.007	0.005	0.005	0.004 $\pm$ 0.001	0.000	0.000
Spy	0.580 $\pm$ 0.039	0.543$\pm$0.036	0.543$\pm$0.036	0.010 $\pm$ 0.001	0.009$\pm$0.001	0.009$\pm$0.001	0.019$\pm$0.001	0.019	0.019

Table S6: Detailed results for `job_offers_pl_corpus` with labelling strategy `mvc` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.575$\pm$0.012	0.667 $\pm$ 0.012	0.798$\pm$0.009	0.178$\pm$0.005	0.221 $\pm$ 0.008	0.355 $\pm$ 0.012	0.000	0.000	0.001 $\pm$ 0.002
Clust	0.345 $\pm$ 0.012	0.587 $\pm$ 0.009	0.663 $\pm$ 0.009	0.120 $\pm$ 0.002	0.195 $\pm$ 0.009	0.256 $\pm$ 0.014	0.287$\pm$0.001	0.040 $\pm$ 0.014	0.090 $\pm$ 0.021
S-LassClust	0.327 $\pm$ 0.006	0.684 $\pm$ 0.012	0.758 $\pm$ 0.013	0.117 $\pm$ 0.001	0.305$\pm$0.020	0.383$\pm$0.024	0.286 $\pm$ 0.001	0.106 $\pm$ 0.021	0.215 $\pm$ 0.019
NS-LassClust	0.325 $\pm$ 0.008	0.684 $\pm$ 0.012	0.758 $\pm$ 0.013	0.117 $\pm$ 0.001	0.305$\pm$0.020	0.383$\pm$0.024	0.286 $\pm$ 0.001	0.106 $\pm$ 0.021	0.215 $\pm$ 0.019
LassoJoint	0.436 $\pm$ 0.011	0.704$\pm$0.019	0.766 $\pm$ 0.013	0.158 $\pm$ 0.001	0.298 $\pm$ 0.020	0.371 $\pm$ 0.019	0.116 $\pm$ 0.016	0.169 $\pm$ 0.023	0.312$\pm$0.017
Spy	0.484 $\pm$ 0.009	0.522 $\pm$ 0.011	0.522 $\pm$ 0.010	0.154 $\pm$ 0.003	0.170 $\pm$ 0.005	0.170 $\pm$ 0.004	0.032 $\pm$ 0.028	0.274$\pm$0.008	0.271 $\pm$ 0.006

Table S7: Detailed results for `open_subtitles_corpus` with labelling strategy `mvc` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.565$\pm$0.012	0.677 $\pm$ 0.011	0.819 $\pm$ 0.012	0.682$\pm$0.011	0.761 $\pm$ 0.012	0.874 $\pm$ 0.012	0.000	0.105 $\pm$ 0.009	0.819 $\pm$ 0.006
Clust	0.532 $\pm$ 0.010	0.488 $\pm$ 0.011	0.492 $\pm$ 0.010	0.633 $\pm$ 0.007	0.604 $\pm$ 0.008	0.606 $\pm$ 0.007	0.357 $\pm$ 0.013	0.751$\pm$0.006	0.777 $\pm$ 0.003
S-LassClust	0.555 $\pm$ 0.013	0.521 $\pm$ 0.011	0.571 $\pm$ 0.010	0.645 $\pm$ 0.009	0.627 $\pm$ 0.008	0.664 $\pm$ 0.009	0.368 $\pm$ 0.037	0.746 $\pm$ 0.006	0.783 $\pm$ 0.004
NS-LassClust	0.558 $\pm$ 0.013	0.521 $\pm$ 0.011	0.571 $\pm$ 0.010	0.648 $\pm$ 0.009	0.627 $\pm$ 0.008	0.664 $\pm$ 0.009	0.387$\pm$0.022	0.746 $\pm$ 0.006	0.783 $\pm$ 0.004
LassoJoint	0.510 $\pm$ 0.012	0.826$\pm$0.014	0.907$\pm$0.008	0.646 $\pm$ 0.006	0.865$\pm$0.014	0.919$\pm$0.009	0.222 $\pm$ 0.043	0.626 $\pm$ 0.012	0.891$\pm$0.006
Spy	0.488 $\pm$ 0.005	0.492 $\pm$ 0.010	0.490 $\pm$ 0.009	0.636 $\pm$ 0.002	0.617 $\pm$ 0.006	0.617 $\pm$ 0.006	0.005 $\pm$ 0.003	0.019 $\pm$ 0.004	0.018 $\pm$ 0.004

Table S8: Detailed results for `plwiki` with labelling strategy `mvc` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.665 $\pm$ 0.010	0.643 $\pm$ 0.010	0.721 $\pm$ 0.008	0.525 $\pm$ 0.009	0.517 $\pm$ 0.009	0.626 $\pm$ 0.013	0.005 $\pm$ 0.002	0.023 $\pm$ 0.005	0.029 $\pm$ 0.008
Clust	0.124 $\pm$ 0.005	0.661 $\pm$ 0.012	0.706 $\pm$ 0.011	0.262 $\pm$ 0.001	0.567 $\pm$ 0.011	0.626 $\pm$ 0.011	0.587 $\pm$ 0.001	0.100 $\pm$ 0.012	0.122 $\pm$ 0.014
S-LassClust	0.153 $\pm$ 0.011	0.688 $\pm$ 0.011	0.733 $\pm$ 0.009	0.266 $\pm$ 0.002	0.598 $\pm$ 0.011	0.662 $\pm$ 0.011	0.588 $\pm$ 0.001	0.301 $\pm$ 0.013	0.348 $\pm$ 0.017
NS-LassClust	0.153 $\pm$ 0.011	0.688 $\pm$ 0.011	0.733 $\pm$ 0.009	0.266 $\pm$ 0.002	0.598 $\pm$ 0.011	0.662 $\pm$ 0.011	0.588 $\pm$ 0.001	0.301 $\pm$ 0.013	0.348 $\pm$ 0.017
LassoJoint	0.729$\pm$0.014	0.781$\pm$0.009	0.853$\pm$0.005	0.590$\pm$0.010	0.727$\pm$0.012	0.812$\pm$0.009	0.691$\pm$0.020	0.620$\pm$0.015	0.643$\pm$0.010
Spy	0.409 $\pm$ 0.022	0.482 $\pm$ 0.005	0.481 $\pm$ 0.005	0.372 $\pm$ 0.011	0.408 $\pm$ 0.003	0.407 $\pm$ 0.002	0.526 $\pm$ 0.019	0.579 $\pm$ 0.003	0.578 $\pm$ 0.003

Table S9: Detailed results for `ulotki_medyczne` with labelling strategy `mvc` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.173 $\pm$ 0.009	0.545 $\pm$ 0.021	0.827 $\pm$ 0.011	0.035	0.065 $\pm$ 0.004	0.243 $\pm$ 0.020	0.000	0.000	0.000
Clust	0.600 $\pm$ 0.012	0.197 $\pm$ 0.007	0.282 $\pm$ 0.011	0.069 $\pm$ 0.002	0.036	0.040 $\pm$ 0.001	0.159 $\pm$ 0.003	0.001 $\pm$ 0.001	0.004 $\pm$ 0.002
S-LassClust	0.677$\pm$0.019	0.262 $\pm$ 0.009	0.331 $\pm$ 0.011	0.089 $\pm$ 0.006	0.039	0.043 $\pm$ 0.001	0.165$\pm$0.003	0.008 $\pm$ 0.002	0.013 $\pm$ 0.004
NS-LassClust	0.676 $\pm$ 0.020	0.262 $\pm$ 0.009	0.331 $\pm$ 0.011	0.089 $\pm$ 0.007	0.039	0.043 $\pm$ 0.001	0.164 $\pm$ 0.003	0.008 $\pm$ 0.002	0.013 $\pm$ 0.004
LassoJoint	0.306 $\pm$ 0.012	0.799 $\pm$ 0.025	0.951$\pm$0.004	0.062	0.193 $\pm$ 0.026	0.522$\pm$0.032	0.001 $\pm$ 0.001	0.000	0.015 $\pm$ 0.007
Spy	0.633 $\pm$ 0.068	0.854$\pm$0.011	0.851 $\pm$ 0.011	0.124$\pm$0.019	0.225$\pm$0.012	0.220 $\pm$ 0.013	0.000	0.159$\pm$0.034	0.149$\pm$0.031

Table S10: Detailed results for `wolne_lektury_corpus` with labelling strategy `mvc` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.626$\pm$0.012	0.650 $\pm$ 0.016	0.718 $\pm$ 0.012	0.416$\pm$0.009	0.431 $\pm$ 0.011	0.517$\pm$0.014	0.000	0.002 $\pm$ 0.002	0.054 $\pm$ 0.013
Clust	0.261 $\pm$ 0.009	0.561 $\pm$ 0.014	0.624 $\pm$ 0.016	0.261 $\pm$ 0.002	0.376 $\pm$ 0.008	0.421 $\pm$ 0.012	0.569$\pm$0.003	0.048 $\pm$ 0.010	0.081 $\pm$ 0.011
S-LassClust	0.298 $\pm$ 0.024	0.596 $\pm$ 0.012	0.656 $\pm$ 0.012	0.270 $\pm$ 0.006	0.398 $\pm$ 0.008	0.438 $\pm$ 0.010	0.569$\pm$0.003	0.098 $\pm$ 0.010	0.168 $\pm$ 0.018
NS-LassClust	0.298 $\pm$ 0.023	0.596 $\pm$ 0.012	0.656 $\pm$ 0.012	0.270 $\pm$ 0.005	0.398 $\pm$ 0.008	0.438 $\pm$ 0.010	0.569$\pm$0.003	0.098 $\pm$ 0.010	0.168 $\pm$ 0.018
LassoJoint	0.493 $\pm$ 0.014	0.660$\pm$0.012	0.726$\pm$0.011	0.375 $\pm$ 0.006	0.445$\pm$0.010	0.499 $\pm$ 0.013	0.251 $\pm$ 0.032	0.157 $\pm$ 0.020	0.262 $\pm$ 0.017
Spy	0.498 $\pm$ 0.012	0.534 $\pm$ 0.007	0.533 $\pm$ 0.007	0.375 $\pm$ 0.006	0.395 $\pm$ 0.004	0.394 $\pm$ 0.004	0.520 $\pm$ 0.012	0.557$\pm$0.004	0.556$\pm$0.004

3 Labelling Strategy: classic

Table S11: Detailed results for ISAP_corpus with labelling strategy `classic` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.826$\pm$0.055	0.757$\pm$0.062	0.645$\pm$0.058	0.038$\pm$0.010	0.025$\pm$0.008	0.014$\pm$0.003	0.000	0.000	0.000
Clust	0.640 $\pm$ 0.092	0.598 $\pm$ 0.086	0.491 $\pm$ 0.062	0.012 $\pm$ 0.003	0.011 $\pm$ 0.003	0.008 $\pm$ 0.001	0.000	0.000	0.000
S-LassClust	0.682 $\pm$ 0.068	0.538 $\pm$ 0.052	0.416 $\pm$ 0.046	0.013 $\pm$ 0.002	0.009 $\pm$ 0.001	0.007 $\pm$ 0.001	0.000	0.000	0.000
NS-LassClust	0.682 $\pm$ 0.068	0.558 $\pm$ 0.091	0.416 $\pm$ 0.046	0.013 $\pm$ 0.002	0.010 $\pm$ 0.002	0.007 $\pm$ 0.001	0.000	0.000	0.000
LassoJoint	0.666 $\pm$ 0.077	0.663 $\pm$ 0.081	0.577 $\pm$ 0.042	0.017 $\pm$ 0.008	0.015 $\pm$ 0.005	0.010 $\pm$ 0.001	0.000	0.000	0.000
Spy	0.543 $\pm$ 0.035	0.543 $\pm$ 0.036	0.543 $\pm$ 0.036	0.009 $\pm$ 0.001	0.009 $\pm$ 0.001	0.009 $\pm$ 0.001	0.019	0.019	0.019

Table S12: Detailed results for job_offers_pl_corpus with labelling strategy `classic` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.822 $\pm$ 0.011	0.828$\pm$0.011	0.819 $\pm$ 0.012	0.377 $\pm$ 0.016	0.384 $\pm$ 0.018	0.373 $\pm$ 0.017	0.000	0.000	0.008 $\pm$ 0.004
Clust	0.662 $\pm$ 0.013	0.713 $\pm$ 0.013	0.730 $\pm$ 0.013	0.267 $\pm$ 0.013	0.309 $\pm$ 0.017	0.322 $\pm$ 0.017	0.039 $\pm$ 0.013	0.068 $\pm$ 0.017	0.117 $\pm$ 0.020
S-LassClust	0.732 $\pm$ 0.015	0.762 $\pm$ 0.037	0.808 $\pm$ 0.013	0.343 $\pm$ 0.016	0.379 $\pm$ 0.052	0.438 $\pm$ 0.023	0.061 $\pm$ 0.015	0.169 $\pm$ 0.099	0.225 $\pm$ 0.017
NS-LassClust	0.732 $\pm$ 0.015	0.781 $\pm$ 0.029	0.808 $\pm$ 0.013	0.343 $\pm$ 0.016	0.400$\pm$0.040	0.438 $\pm$ 0.023	0.061 $\pm$ 0.015	0.088 $\pm$ 0.040	0.225 $\pm$ 0.017
LassoJoint	0.892$\pm$0.008	0.803 $\pm$ 0.026	0.888$\pm$0.009	0.635$\pm$0.021	0.398 $\pm$ 0.028	0.632$\pm$0.020	0.590$\pm$0.013	0.233 $\pm$ 0.201	0.577$\pm$0.017
Spy	0.521 $\pm$ 0.010	0.521 $\pm$ 0.010	0.521 $\pm$ 0.010	0.170 $\pm$ 0.004	0.169 $\pm$ 0.004	0.169 $\pm$ 0.004	0.271 $\pm$ 0.006	0.268$\pm$0.005	0.267 $\pm$ 0.006

Table S13: Detailed results for open_subtitles_corpus with labelling strategy `classic` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.704$\pm$0.023	0.852$\pm$0.016	0.856$\pm$0.010	0.786$\pm$0.019	0.893$\pm$0.012	0.892$\pm$0.009	0.004 $\pm$ 0.002	0.131 $\pm$ 0.028	0.853$\pm$0.005
Clust	0.482 $\pm$ 0.011	0.486 $\pm$ 0.010	0.490 $\pm$ 0.010	0.600 $\pm$ 0.007	0.603 $\pm$ 0.007	0.606 $\pm$ 0.007	0.741 $\pm$ 0.006	0.772 $\pm$ 0.003	0.780 $\pm$ 0.002
S-LassClust	0.502 $\pm$ 0.012	0.518 $\pm$ 0.026	0.582 $\pm$ 0.009	0.612 $\pm$ 0.009	0.623 $\pm$ 0.018	0.672 $\pm$ 0.010	0.739 $\pm$ 0.005	0.776$\pm$0.005	0.785 $\pm$ 0.002
NS-LassClust	0.502 $\pm$ 0.012	0.531 $\pm$ 0.013	0.582 $\pm$ 0.009	0.612 $\pm$ 0.009	0.633 $\pm$ 0.011	0.672 $\pm$ 0.010	0.739 $\pm$ 0.005	0.713 $\pm$ 0.163	0.785 $\pm$ 0.002
LassoJoint	0.469 $\pm$ 0.014	0.846 $\pm$ 0.119	0.508 $\pm$ 0.015	0.604 $\pm$ 0.016	0.879 $\pm$ 0.082	0.641 $\pm$ 0.009	0.788$\pm$0.002	0.754 $\pm$ 0.192	0.789 $\pm$ 0.001
Spy	0.490 $\pm$ 0.010	0.489 $\pm$ 0.009	0.488 $\pm$ 0.009	0.617 $\pm$ 0.006	0.617 $\pm$ 0.006	0.617 $\pm$ 0.006	0.018 $\pm$ 0.004	0.018 $\pm$ 0.004	0.018 $\pm$ 0.004

Table S14: Detailed results for plwiki with labelling strategy `classic` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.722 $\pm$ 0.023	0.730 $\pm$ 0.012	0.731 $\pm$ 0.009	0.587 $\pm$ 0.035	0.607 $\pm$ 0.020	0.622 $\pm$ 0.016	0.000	0.000 $\pm$ 0.001	0.004 $\pm$ 0.004
Clust	0.710 $\pm$ 0.023	0.705 $\pm$ 0.016	0.720 $\pm$ 0.011	0.654 $\pm$ 0.027	0.638 $\pm$ 0.020	0.648 $\pm$ 0.015	0.011 $\pm$ 0.005	0.019 $\pm$ 0.007	0.045 $\pm$ 0.011
S-LassClust	0.429 $\pm$ 0.034	0.674 $\pm$ 0.050	0.718 $\pm$ 0.011	0.416 $\pm$ 0.027	0.615 $\pm$ 0.055	0.652 $\pm$ 0.015	0.018 $\pm$ 0.007	0.046 $\pm$ 0.023	0.117 $\pm$ 0.015
NS-LassClust	0.429 $\pm$ 0.034	0.643 $\pm$ 0.044	0.718 $\pm$ 0.011	0.416 $\pm$ 0.027	0.577 $\pm$ 0.048	0.652 $\pm$ 0.015	0.018 $\pm$ 0.007	0.030 $\pm$ 0.013	0.117 $\pm$ 0.015
LassoJoint	0.869$\pm$0.010	0.790$\pm$0.065	0.881$\pm$0.006	0.841$\pm$0.013	0.718$\pm$0.105	0.851$\pm$0.010	0.527 $\pm$ 0.054	0.192 $\pm$ 0.165	0.754$\pm$0.011
Spy	0.481 $\pm$ 0.005	0.480 $\pm$ 0.005	0.480 $\pm$ 0.005	0.407 $\pm$ 0.002	0.407 $\pm$ 0.002	0.407 $\pm$ 0.002	0.578$\pm$0.003	0.578$\pm$0.003	0.578 $\pm$ 0.003

Table S15: Detailed results for `ulotki_medyczne` with labelling strategy `classic` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.920 $\pm$ 0.004	0.917 $\pm$ 0.005	0.915 $\pm$ 0.006	0.421 $\pm$ 0.022	0.413 $\pm$ 0.023	0.406 $\pm$ 0.024	0.000	0.000	0.005 $\pm$ 0.008
Clust	0.381 $\pm$ 0.012	0.427 $\pm$ 0.012	0.457 $\pm$ 0.010	0.046 $\pm$ 0.001	0.050 $\pm$ 0.001	0.052 $\pm$ 0.001	0.012 $\pm$ 0.002	0.016 $\pm$ 0.003	0.022 $\pm$ 0.005
S-LassClust	0.369 $\pm$ 0.013	0.424 $\pm$ 0.020	0.462 $\pm$ 0.010	0.045 $\pm$ 0.001	0.049 $\pm$ 0.002	0.053 $\pm$ 0.001	0.018 $\pm$ 0.004	0.040 $\pm$ 0.017	0.042 $\pm$ 0.006
NS-LassClust	0.369 $\pm$ 0.013	0.438 $\pm$ 0.017	0.462 $\pm$ 0.010	0.045 $\pm$ 0.001	0.051 $\pm$ 0.002	0.053 $\pm$ 0.001	0.018 $\pm$ 0.004	0.022 $\pm$ 0.009	0.042 $\pm$ 0.006
LassoJoint	0.989$\pm$0.002	0.946$\pm$0.049	0.990$\pm$0.002	0.782$\pm$0.041	0.578$\pm$0.185	0.811$\pm$0.040	0.782$\pm$0.021	0.420$\pm$0.358	0.793$\pm$0.020
Spy	0.849 $\pm$ 0.011	0.847 $\pm$ 0.011	0.845 $\pm$ 0.011	0.217 $\pm$ 0.011	0.214 $\pm$ 0.011	0.211 $\pm$ 0.010	0.148 $\pm$ 0.030	0.145 $\pm$ 0.032	0.141 $\pm$ 0.033

Table S16: Detailed results for `wolne_lektury_corpus` with labelling strategy `classic` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.693 $\pm$ 0.015	0.729$\pm$0.013	0.748 $\pm$ 0.012	0.478 $\pm$ 0.015	0.523$\pm$0.017	0.555$\pm$0.018	0.000	0.003 $\pm$ 0.002	0.327 $\pm$ 0.030
Clust	0.589 $\pm$ 0.023	0.643 $\pm$ 0.017	0.666 $\pm$ 0.016	0.406 $\pm$ 0.017	0.437 $\pm$ 0.015	0.456 $\pm$ 0.015	0.036 $\pm$ 0.009	0.058 $\pm$ 0.013	0.172 $\pm$ 0.026
S-LassClust	0.438 $\pm$ 0.033	0.621 $\pm$ 0.019	0.664 $\pm$ 0.013	0.320 $\pm$ 0.014	0.425 $\pm$ 0.015	0.458 $\pm$ 0.012	0.046 $\pm$ 0.011	0.090 $\pm$ 0.027	0.279 $\pm$ 0.028
NS-LassClust	0.438 $\pm$ 0.033	0.621 $\pm$ 0.014	0.664 $\pm$ 0.013	0.320 $\pm$ 0.014	0.424 $\pm$ 0.012	0.458 $\pm$ 0.012	0.046 $\pm$ 0.011	0.078 $\pm$ 0.020	0.279 $\pm$ 0.028
LassoJoint	0.767$\pm$0.009	0.709 $\pm$ 0.029	0.776$\pm$0.011	0.544$\pm$0.013	0.488 $\pm$ 0.027	0.554 $\pm$ 0.015	0.601$\pm$0.028	0.299 $\pm$ 0.251	0.681$\pm$0.009
Spy	0.533 $\pm$ 0.007	0.534 $\pm$ 0.007	0.533 $\pm$ 0.007	0.394 $\pm$ 0.004	0.394 $\pm$ 0.004	0.394 $\pm$ 0.004	0.556 $\pm$ 0.004	0.556$\pm$0.004	0.556 $\pm$ 0.004

4 Feature Descriptions

Table S17 provides a comprehensive list of all 22 linguistic and stylometric features used in the benchmark. All numeric features are rescaled to the $[0, 1]$ range using min-max normalization before model training.

Table S17: Description of 22 input features (X) used in the benchmark.

Feature	Description
<code>adj_ratio</code>	Ratio of adjectives to total words
<code>adjectives</code>	Count of adjectives in the document
<code>adverbs</code>	Count of adverbs in the document
<code>avg_sentence_length</code>	Average number of words per sentence
<code>avg_word_length</code>	Average number of characters per word
<code>camel_case</code>	Count of camelCase words
<code>capitalized_words</code>	Count of words starting with uppercase letter
<code>characters</code>	Total character count (excluding whitespace)
<code>gunning_fog</code>	Gunning Fog index; estimates U.S. school grade needed to understand text
<code>lexical_density</code>	Ratio of content words (nouns, verbs, adjectives, adverbs) to total words
<code>noun_ratio</code>	Ratio of nouns to total words
<code>nouns</code>	Count of nouns in the document
<code>oovs</code>	Count of out-of-vocabulary words (not in standard Polish dictionary)
<code>pos_num</code>	Count of numerals (number-tagged part-of-speech tokens)
<code>pos_x</code>	Count of unknown or rare part-of-speech categories
<code>punctuations</code>	Count of punctuation marks
<code>sentences</code>	Count of sentences (detected by sentence boundary markers)
<code>stopwords</code>	Count of common Polish stopwords (high-frequency, low-content words)
<code>symbols</code>	Count of non-alphanumeric symbols (special characters)
<code>verb_ratio</code>	Ratio of verbs to total words
<code>verbs</code>	Count of verbs in the document
<code>words</code>	Total word count

5 Statistical Comparison for F1, AUC, AUCPR for different PU methods with classic and MVC scenarios

Table S18: Results of the paired Wilcoxon signed-rank tests comparing the classic and non-SCAR variants based on the F1 score. W – the test statistics. For each threshold parameter c , Holm’s correction was applied across the six tested methods. Statistically significant adjusted p -values ($\alpha = 0.05$) are shown in bold.

Method	$c = 0.2$		$c = 0.5$		$c = 0.8$	
	W	Holm p	W	Holm p	W	Holm p
Clust	1475	1.51×10^{-4}	445	3.87×10^{-1}	376	5.67×10^{-2}
LassoJoint	210	1.07×10^{-6}	447	4.49×10^{-1}	79.5	4.42×10^{-7}
Naive	120	5.86×10^{-1}	191	1.00	196.5	3.27×10^{-3}
Non-strict LassClust	1475	1.51×10^{-4}	878.5	1.04×10^{-1}	486.5	4.24×10^{-1}
Spy	21	1.43×10^{-9}	801	8.39×10^{-7}	422	4.49×10^{-5}
Strict LassClust	1475	1.51×10^{-4}	586.5	1.00	486.5	4.24×10^{-1}

Table S19: Results of the paired Wilcoxon signed-rank tests comparing the classic and non-SCAR variants based on the AUCPR score. W – the test statistics. For each threshold parameter c , Holm’s correction was applied across the six tested methods. Statistically significant adjusted p -values ($\alpha = 0.05$) are shown in bold.

Method	$c = 0.2$		$c = 0.5$		$c = 0.8$	
	W	Holm p	W	Holm p	W	Holm p
Clust	465	2.81×10^{-3}	45	1.16×10^{-9}	7	6.66×10^{-10}
LassoJoint	152	9.93×10^{-8}	313	2.85×10^{-5}	460	8.20×10^{-4}
Naive	0	1.00×10^{-10}	0	1.00×10^{-10}	114	3.01×10^{-8}
Non-strict LassClust	540	1.07×10^{-2}	338.5	4.32×10^{-5}	246	3.42×10^{-6}
Spy	277	5.14×10^{-5}	402	5.77×10^{-5}	235	7.45×10^{-4}
Strict LassClust	536	1.07×10^{-2}	279	1.16×10^{-5}	246	3.42×10^{-6}

Table S20: Results of the paired Wilcoxon signed-rank tests comparing the classic and non-SCAR variants based on the AUC score. W – the test statistics. For each threshold parameter c , Holm’s correction was applied across the six tested methods. Statistically significant adjusted p -values ($\alpha = 0.05$) are shown in bold.

Method	$c = 0.2$		$c = 0.5$		$c = 0.8$	
	W	Holm p	W	Holm p	W	Holm p
Clust	464	2.74×10^{-3}	45	1.17×10^{-9}	33.5	1.00×10^{-9}
LassoJoint	55	1.25×10^{-9}	189	2.77×10^{-7}	546	6.67×10^{-3}
Naive	0	1.00×10^{-10}	0	1.00×10^{-10}	4	1.23×10^{-10}
Non-strict LassClust	699	2.12×10^{-1}	222.5	8.06×10^{-7}	188	5.61×10^{-7}
Spy	289	1.65×10^{-5}	1036	2.05×10^{-7}	1012	5.61×10^{-7}
Strict LassClust	695	2.12×10^{-1}	200.5	8.06×10^{-7}	187	5.61×10^{-7}